

variables
originales

Comunalidad

Unicidad

$Cov(F_i, F_j) = 0$

$$\begin{aligned}
 x_1 &= 0.1F_1 + 0.2F_2 + 0.5F_3 + 0.2 \\
 x_2 &= 0.7F_1 + 0.1F_2 + 0.2 \\
 x_3 &= 0.1F_1 + 0.5F_2 + 0.1F_3 + 0.3 \\
 x_4 &= 0.2F_1 + 0.3F_2 + 0.3F_3 + 0.2 \\
 x_5 &= 0.1F_1 + 0.2F_2 + 0.7 \\
 x_6 &= 0.1F_2 + 0.8F_3 + 0.1
 \end{aligned}$$

F_1 F_2 F_3
 x_1 x_2 x_3
 x_4 x_5 x_6

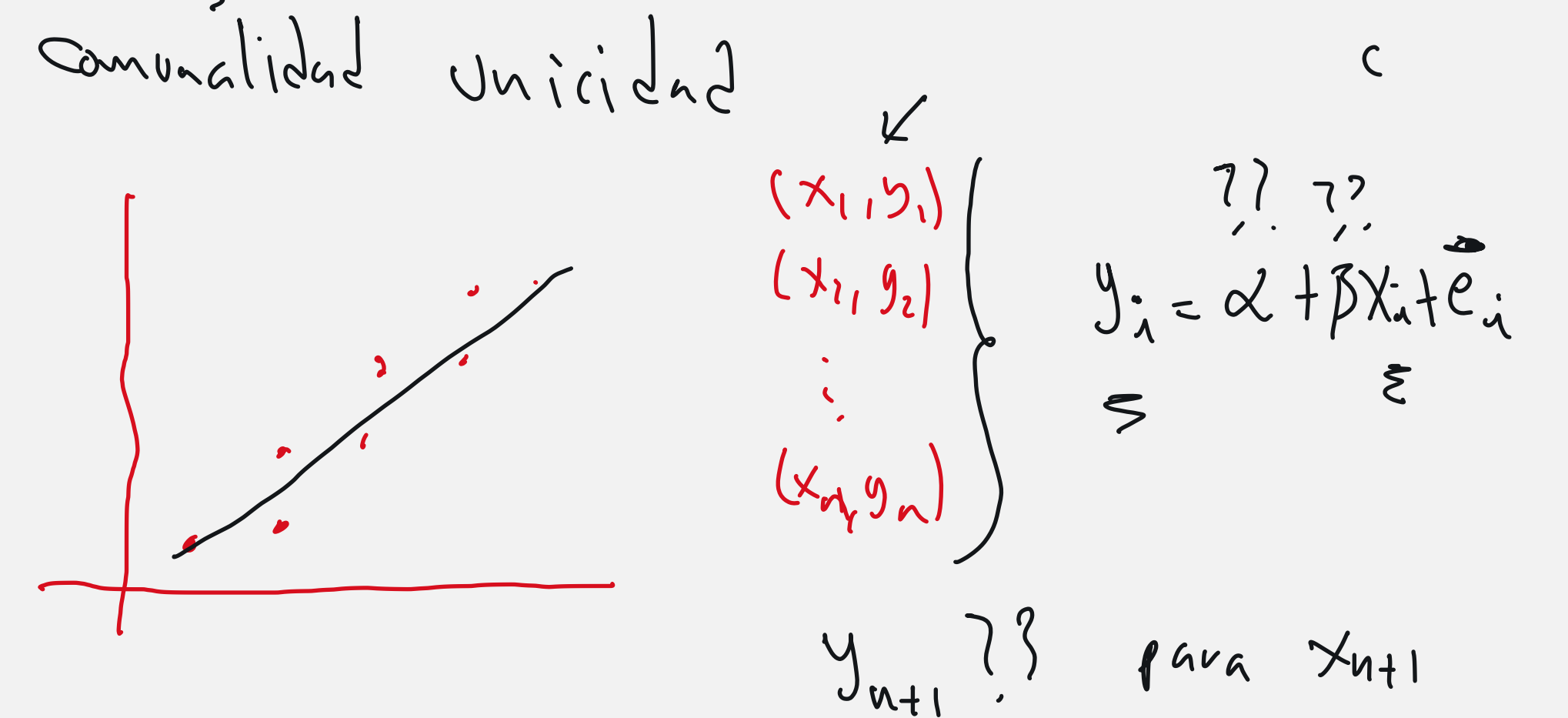
$$\begin{aligned}
 X &= \mu + AF + U \\
 \rightarrow Cov(F, U) &= 0 \quad Cov(F) = I \quad Cov(U) = \Psi \quad E(U) = 0 \\
 \Sigma_x &= E(X - \mu)(X - \mu)^t = E(\mu + AF + U - \mu)(X - \mu)^t \\
 &= E[(AF + U)(AF + U)^t] = E(AF(AF)^t + E(AFU^t) + E(UU^t)) \\
 &= E(AF(F^t A^t) + A E(FU^t) + \Psi) \\
 &= A E(F F^t) A^t + A(0) + \Psi \\
 &= A I A^t + 0 + \Psi = \underline{A A^t + \Psi}
 \end{aligned}$$

$X = AF + U$
 $F = Z X \leftarrow ??$
 $Y = X\beta + e$
 $\hat{\beta} = (X^t X)^{-1} X^t Y$

Regres Y
 X
 β
 e

Fact. X
 A
 F
 U

$\hat{F} = (A^t A)^{-1} A^t X$



$$\begin{aligned}
 e_i &= y_i - \alpha - \beta x_i \quad e_i^2 = (y_i - \alpha - \beta x_i)^2 \\
 \Sigma e_i^2 &= \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2 \leftarrow \text{Minimos cuadrados}
 \end{aligned}$$

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + e_i$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \quad X = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

$$e^t = (Y - X\beta)^t (Y - X\beta)$$

$$\hat{\beta} = (X^t X)^{-1} X^t Y$$